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From data to dignity: Understanding and predicting fuel poverty in the United Kingdom with machine learning

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ABSTRACT

This study applies a data-driven framework to assess fuel poverty risk in England using the 2019 English Housing Survey. A clustering-prediction-interpretation approach combines k-prototypes clustering, XGBoost modeling, and Shapley Additive Explanations (SHAP) to identify household vulnerability profiles and the key drivers of fuel poverty. The study segments the entire dataset into three clusters: households with moderate energy costs, younger employed households with high appliance and lighting costs, and older low-income economically inactive households, who emerge as the most vulnerable group. Four XGBoost models—one for the entire dataset and three cluster-specific models—were developed to predict household fuel poverty. Predictive performance was high across models, and SHAP analysis revealed that the influence of socio-economic, housing, and energy-lighting cost features varies significantly across clusters. The results demonstrate that fuel poverty is heterogeneous and cannot be effectively addressed by uniform national schemes. Instead, differentiated interventions—such as targeted financial transfers for vulnerable households, retrofitting programs, and appliance-efficiency policies—are required. Methodologically, the study highlights the value of integrating clustering with predictive modeling to uncover hidden vulnerability patterns that conventional static indicators often overlook. While the analysis focuses on England and uses cross-sectional data, the proposed framework is adaptable to other contexts and can be extended with longitudinal data to strengthen generalizability.

1. Introduction

The ongoing cost-of-living crisis and successive energy price shocks have sharply intensified the challenge of fuel poverty in England [1]. In 2023, an estimated 3.17 million households—around 13 % of all households—were officially classified as fuel-poor under the Low-Income Low-Energy-Efficiency (LILEE) indicator [2]. For these households, escaping fuel poverty would require reducing annual energy costs by an average of £417, with the shortfall rising to more than £780 in rural areas and among households off the gas grid [2]. Such figures highlight not only the scale of the problem but also its deeply uneven distribution, with the greatest burdens falling on those already structurally disadvantaged.

Fuel poverty emerges from the intersection of multiple factors: persistently low incomes, an inefficient and aging housing stock, rising energy costs, and household characteristics such as age, disability, or

health status. Older residents and people with disabilities, for example, often require above-average energy consumption for heating or medical needs, yet lack the resources to retrofit their homes [3]. Low-income families face equally stark trade-offs between meeting energy needs and other essentials, with lasting effects on children's health, educational attainment, and social mobility [4]. More recent evidence shows that the energy price crisis has disproportionately impacted the lowest income quintiles, deepening inequalities and exposing households to sustained financial strain [5,6].

Successive governments have attempted to mitigate fuel poverty through initiatives like the Warm Home Discount and the Great British Insulation Scheme. Although insulation upgrades can lower household energy bills by £300–400 annually, evaluations reveal that such interventions often miss the most vulnerable populations, leaving many outside the scope of these schemes [7]. Additionally, retrofit policies frequently focus only on the technical dimension of housing efficiency, without addressing compounding factors such as low income, household

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List of abbreviations and notations

AFCP	After Fuel Cost Poverty
CPU	Central Processing Unit
DW	Dwelling
EDS	Entire Dataset
EHS	English Housing Survey
EPOV	Energy Poverty Observatory
EU	European Union
GBDT	Gradient Boosting Decision Tree
HH	Household
IAMs	Integrated Assessment Models
LIHC	Low-Income High Costs
MHCLG	Ministry of Housing, Communities, and Local Government
ML	Machine Learning
SHAP	Shapley Additive Explanations
SVM	Support Vector Machine
XGBoost	Extreme Gradient Boosting
$Cov(X,Y)$	Covariance between feature X and feature Y
$E(X)$	Expected value of feature X
$E(Y)$	Expected Value of feature Y
$E(X^2)$	Expected value of the squared of feature X
$E(Y^2)$	Expected value of the squared of feature Y

E_L^n	Total cost related to numerical features
E_L^c	Total cost related to categorical features
E_L	Combined cost function (numerical and categorical)
$f_t(x_i)$	New generated tree model
K	Total number of prototypes
L	Index for prototype l
m	Total number of features
$m-p$	Total number of categorical features
p	Total number of numerical features
Q_l	Center of prototype l
q_{lj}^c	The value of the j th categorical feature at the center of prototype l
q_{lj}^n	The value of the j th numerical feature at the center of prototype l
T	Number of leaves
$w_{i,l}$	Dummy variable for the membership of data point i to prototype l
X_{ij}^c	The value of the j th categorical feature of the i th data point
X_{ij}^n	The value of the j th numerical feature of the i th data point
y_l	Weight for the categorical features of cluster l
$\hat{y}_i^{(t-1)}$	Previous generated tree model
$\hat{y}_i^{(t)}$	Final tree model

composition, or dwelling type—limitations that restrict their transformative potential [8]. This disjunction between policy scope and household realities highlights the urgent need for refined tools that can identify at-risk groups and shape more targeted, cost-effective interventions.

Academic discourse on fuel poverty measurement has also evolved considerably. Boardman's original 10 % expenditure threshold conceptualized fuel poverty primarily as an affordability issue [9]. Later metrics such as the Low-Income High-Cost (LIHC) and the current LILEE standard sought to capture the joint influence of income and housing energy performance. Yet these static measures remain limited: they provide headline prevalence statistics but fail to reflect the heterogeneity of household vulnerabilities or the dynamic conditions that drive them [10,11]. As critics argue, vulnerability is socially and spatially embedded, shaped by structural inequalities as well as individual circumstances, and thus not easily captured by uniform thresholds.

To address these shortcomings, recent research has turned to data-driven approaches. Clustering and machine learning (ML) methods offer a way to detect hidden patterns in socio-economic and housing data that conventional indicators often obscure. Zheng et al. [12] used English Housing Survey (EHS) data to train predictive models that outperformed government benchmarks in identifying households at risk of fuel poverty, while Al Kez [13] combined socio-economic and remote sensing data to map risk at a fine-grained spatial scale. These advances underscore the potential of ML, but they also reveal an important gap: few studies combine clustering, predictive modeling, and interpretability to simultaneously identify distinct vulnerability profiles, assess predictive accuracy across them, and generate policy-relevant insights that clarify not only who is at risk but also why.

This paper contributes to filling that gap. Using the English Housing Survey, we develop a clustering-based predictive framework that combines k-prototypes clustering, XGBoost modeling, and SHAP interpretability. Specifically, k-prototypes groups households into distinct vulnerability profiles, XGBoost predicts fuel poverty risk within these clusters, and SHAP explains the relative contribution of socio-economic, housing, and energy-cost features to each prediction. This integrated approach goes beyond treating households as a homogeneous group, instead revealing heterogeneous vulnerability pathways and offering interpretable insights into the drivers of risk. Importantly, the emphasis

on interpretability ensures that the results do not remain abstract statistical outputs but instead inform actionable strategies for policy-makers. By clarifying both the extent of fuel poverty and the reasons households fall into it, the framework supports the design of interventions that are better tailored to the needs of specific household segments.

Accordingly, this study asks three guiding questions. First, which socio-economic, housing, and energy-related features most strongly explain the likelihood of fuel poverty in England? Second, how can clustering methods enhance predictive models by revealing distinct vulnerability profiles that conventional indicators overlook? And third, what policy lessons emerge from these profiles, particularly in terms of designing more effective and equitable interventions to alleviate fuel poverty? Together, these questions reflect an effort not only to refine measurement but also to strengthen the link between diagnosis and policy design.

The remainder of the paper is organized as follows. Section 2 provides some related studies to the fuel poverty measurement and prediction techniques. Section 3 describes the applied method and dataset. Section 4 presents the modeling results. Section 5 concludes and gives the policy implications.

2. Literature review

Ensuring access to affordable and adequately heated housing is a fundamental principle of sustainable energy policy. However, many households remain unable to achieve “affordable warmth” due to the combined effects of low income, energy-inefficient dwellings, and rising energy prices. Such households are typically classified as being in fuel poverty, historically defined in the UK as requiring 10 % of their income to be spent on energy to achieve adequate thermal comfort [14]. Fuel poverty is not only an economic issue but also a social one, with well-documented adverse impacts on physical and mental health, educational outcomes, and overall quality of life [15]. Recent studies further emphasize the links between energy deprivation, vulnerability, and wider social inequalities [15]. Against this backdrop, the literature review serves two purposes: first, to examine the evolution of measurement indicators in the UK and their limitations; and second, to assess the emergence of predictive and profiling approaches that move beyond

static metrics.

2.1. Fuel poverty measurement in the UK

Indicators of fuel poverty have played a central role in defining the scale of the problem and shaping policy responses in the UK. Over the past three decades, approaches have evolved substantially. The earliest official definition was the “10 % rule”, which classified households as fuel-poor if they needed to spend more than 10 % of income on adequate heating [9]. This was later replaced by the Low-Income High-Costs (LIHC) indicator, which defined households as fuel-poor when their required energy costs were above the national median and their residual income fell below the poverty line [16,17].

Since 2021, the statutory measure in England has been the Low-Income Low-Energy-Efficiency (LILEE) indicator, which classifies a household as fuel-poor if it resides in a property with a Fuel Poverty Energy Efficiency Rating (FPEER)—derived from its Energy Performance Certificate (EPC)—of band D or below and has residual income below the official poverty threshold [18,19]. While this alignment with housing efficiency marks a policy advancement, LILEE remains a static measure that fails to capture heterogeneity in household vulnerabilities such as exposure to energy price shocks or specific health and occupancy-related needs [10,11]. Furthermore, recent empirical studies have reinforced these critiques. Semple [20] provides a comprehensive review showing that existing definitions, including LILEE, systematically undercount households in energy deprivation, particularly in urban areas where cost burdens diverge sharply from national averages. Longhurst and Hargreaves [21] demonstrate how local authority-level data reveal strong spatial clustering of fuel poverty across England, further exposing how static indicators can mask geographic inequalities. Hihetah et al. [22], in a systematic review of lived experiences, argue that focusing only on income and efficiency neglects critical non-monetary dimensions such as disability, heating adequacy, and coping strategies. Willand et al. [23] likewise reframe energy needs in terms of essential “functionings”, underscoring the inadequacy of uniform thresholds for capturing diverse household realities.

Thus, while official indicators such as LIHC and LILEE remain indispensable for statutory monitoring, they are blunt instruments for understanding the full complexity of fuel poverty. Their static design provides headline prevalence statistics but fails to capture heterogeneity in household vulnerabilities or anticipate emerging risks during crises. This has led to growing calls for complementary approaches that move beyond prevalence counts to profile households and identify dynamic risk factors [24,25]. Addressing these gaps is crucial to the present study, which advances a predictive and clustering-based framework designed to uncover vulnerability profiles and provide policy-relevant insights.

2.2. Fuel poverty prediction techniques and clustering

In response to the limitations of static indicators, research has increasingly turned to predictive and profiling approaches that can better capture the heterogeneity of fuel poverty. Early work relied on regression-based and econometric models, which identified statistical associations between income, housing characteristics, and fuel poverty incidence [26]. While these methods clarified structural drivers, they were constrained by relatively small sets of variables and could not easily account for behavioral or health-related dynamics. Spatial econometric and interpolation techniques, such as kriging, expanded analysis by estimating prevalence in data-scarce areas [27,28]. Although useful for mapping broad spatial patterns, these approaches often introduced assumptions that oversimplified the heterogeneity of local experiences. Extensions, such as gender-sensitive vulnerability mapping in England [29], highlighted how fuel poverty interacts with demographic and socio-spatial inequalities, but also illustrates the methodological challenges of capturing these dimensions in practice.

More recently, attention has shifted to clustering and profiling methods that reveal the diversity of fuel poverty experiences. Papada and Kaliampakos [30] applied composite indicators and clustering to examine household energy deprivation in Greece, while Castaño-Rosa et al. [10] used multidimensional clustering to explore heterogeneous vulnerability patterns across EU households. In the English context, Al Kez et al. [13] developed a machine-learning-based profiling framework using the English Housing Survey and the LILEE metric, showing how clustering can reveal hidden vulnerability pathways. Robinson et al. [31] demonstrated how overlapping domestic and transport energy vulnerabilities create distinctive multi-domain risk profiles at the local level. Together, such work underscores that fuel poverty is not monolithic but stratified across household types, requiring tools that differentiate vulnerability segments rather than treating all fuel-poor households as a uniform group.

The emergence of ML has further expanded the scope of prediction and profiling by enabling analysis of large, complex datasets and capturing non-linear relationships among predictors. Studies have demonstrated its value in diverse contexts: Kontokosta and Tull [32] predicted building-level energy burdens in New York City; Xu et al. [27] integrated remote sensing and socio-economic data to estimate poverty incidence at fine spatial scales; and Zheng and McKenna [33] showed that ML models such as XGBoost significantly outperformed government benchmarks in identifying at-risk households using English Housing Survey data. ML approaches are typically divided into supervised algorithms (e.g. random forests, SVMs, neural networks) that classify outcomes, and unsupervised methods (e.g. k-means, k-prototypes) that cluster households based on shared characteristics. Integrating these approaches allows both prediction of fuel poverty risk and discovery of hidden vulnerability profiles.

Alongside technical innovation, recent research has emphasized that methodological rigor and transparency are essential if predictive approaches are to inform effective fuel poverty policy. Ramsden [34] showed that retrofit and advice schemes systematically miss certain groups, illustrating the need to identify not only who benefits but also who remains excluded. Kashour et al. [35] likewise highlight that weak or inconsistent definitions of energy poverty risk are misdirecting interventions, underscoring the importance of robust and transparent measurement. Building on these insights, recent studies stress that predictive models must not only achieve high accuracy but also offer interpretable outputs that explain vulnerability pathways. Without this balance, advanced tools risk becoming black boxes that are difficult to translate into policy action.

Taken together, these contributions mark a clear trajectory: from regression and mapping methods toward more sophisticated, data-driven approaches that combine prediction with profiling. Yet significant gaps remain. Most studies focus either on prevalence or on prediction accuracy, without integrating clustering, predictive modeling, and interpretability in a single framework. This study responds directly to that gap by applying an unsupervised clustering algorithm to segment English households into vulnerability profiles and then using XGBoost with SHAP to predict risk and explain its drivers within each cluster.

Compared to traditional econometric and mapping approaches, which often oversimplify household heterogeneity or fail to account for non-linear relationships, our framework combines k-prototypes, XGBoost, and SHAP to overcome these limitations. Clustering captures the diversity of socio-economic and housing conditions, predictive modeling provides robust accuracy, and interpretability methods reveal the underlying drivers of risk. This integrated design extends beyond prior approaches, linking methodological rigor with actionable insights for policy.

3. Data and methodology used

This section introduces the dataset (3.1) and outlines the methodological steps (3.2): preprocessing, feature selection, clustering,

prediction, and interpretability. Detailed mathematical derivations are provided in [Appendix A.1 \(Table A1\)](#).

3.1. Data

Our study draws on the English Housing Survey (EHS): Fuel Poverty Dataset (2019), a nationally representative survey commissioned by the Ministry of Housing, Communities and Local Government (MHCLG). The dataset is part of the continuous EHS, which collects detailed information on household living conditions and the energy efficiency of dwellings [36]. The fuel poverty module covers the period April 2018–March 2020, with April 2019 as the midpoint, and contains information on 11,974 households. Variables include household income, dwelling type, tenure, heating and insulation systems, and disaggregated energy costs.

Although 2019 is the most recent release of the fuel poverty module available at the time of analysis, it remains widely used in empirical research due to its coverage of both socio-economic and building-level energy efficiency indicators. The dataset is publicly accessible via the UK Data Service. All data are available on the UK Data Service, a national data archive that provides research access to a range of social and economic data collections [37].

[Table 1](#) summarizes the variables selected for analysis, grouped into three categories: energy costs, housing characteristics, and socio-economic attributes, based on prior literature on fuel poverty determinants and data availability. This categorization reflects the established understanding that fuel poverty arises from the interaction of high energy needs, inefficient housing, and limited household resources [16,38,39]. Energy cost variables (e.g., space heating, lighting, cooking, water heating) directly capture affordability pressures and are standard indicators of household energy burden in the UK studies [40,41]. Housing variables such as tenure type, dwelling type, dwelling age,

insulation, and heating system represent structural conditions that shape energy efficiency, with older and poorly insulated homes consistently identified as drivers of fuel poverty [10,13]. Socio-economic variables, including income, household composition, age of the oldest member, employment status, and health status, capture vulnerability pathways beyond physical housing conditions. Prior research shows that low income and unemployment are key drivers of fuel poverty [16], while the presence of long-term illness or disability significantly increases energy needs and the risk of deprivation [15]. Similarly, dwelling age is used in the UK studies as a proxy for housing stock efficiency, with older properties exhibiting systematically lower energy efficiency and higher risk of fuel poverty [42].

While these variables were primarily selected on the basis of established theoretical and empirical research, we also applied Pearson correlation analysis as a supplementary step to avoid redundancy among closely related features.

To illustrate sample characteristics, we also computed descriptive statistics (see [Appendix A.1](#)), reporting mean values for continuous variables (e.g., household income, energy costs) and percentage distributions for categorical variables (e.g., tenure, heating system, insulation).

3.2. Methodology

The analytical framework proceeds in four stages, as summarized in [Fig. 1](#). After cleaning and preprocessing the dataset, relevant features were selected based on prior literature and correlation screening. Households were then clustered into vulnerability groups using the k-prototypes algorithm, after which an XGBoost model was applied to predict fuel poverty risk within each cluster. Finally, model explainability was achieved through SHAP values, and performance was evaluated using standard classification metrics.

3.2.1. Data preprocessing and feature selection

The dataset was first checked to be clean, and the missing data were dropped. Given the mix of categorical and numerical variables, categorical features were converted to numerical format using one-hot encoding, which is a method used to handle categorical data without an ordinal relationship [43]. In this method, each category is converted into a new binary column, where the presence of a category is designated as one (true) or zero (false). This reformatted data is then suitable for use with ML algorithms in predictive modeling.

After encoding the categorical variables, the dataset is ready for the normalization process. The normalization process ensures that all data is scaled to the same range, typically from 0 to 1. This process helps to make the various features comparable and improves the performance of ML algorithms; for this purpose, the min-max normalization equation is expressed as follows:

$$x_N = \frac{x - x_{\min}}{x_{\max} - x_{\min}}, \quad (1)$$

where x is a value of the dataset, x_N is a normalized value, and $x_{\max}$ and $x_{\min}$ are the maximum and the minimum values in the dataset.

Feature selection was guided by two criteria: (i) theoretical relevance from prior studies on fuel poverty determinants, and (ii) correlation screening to avoid redundancy. Pearson correlation coefficients were computed for all variables with a conservative threshold of 0.6 to identify potential multicollinearity [44]. As shown in [Appendix A.2 \(Fig. A1\)](#), the heatmaps indicate that most variables were weakly correlated. An exception was “CookCost” and “WaterHeatCost”, which showed moderate correlation; both were retained because they capture distinct household expenditure categories (cooking versus water heating). Categorical variables, after one-hot encoding, showed generally low correlations and were therefore all maintained.

Table 1
Variables used for multi-dimensional fuel poverty analysis.

Category	Variables	Type	Description
Energy costs	SpaceHeatingCost	Numerical	Annual expenditure on space heating
	LightsCost	Numerical	Annual expenditure on lighting and appliances
	CookCost	Numerical	Annual expenditure on cooking energy
	WaterHeatCost	Numerical	Annual expenditure on water heating
Housing characteristics	Tenuretype	Categorical	Tenure of the dwelling (owner-occupied, rented, etc.)
	HHcomposition	Categorical	Household composition (single, couple, family, etc.)
	Loftins	Categorical	Presence of loft insulation
	FuelPoverty	Categorical	Fuel poverty flag (LILEE indicator)
	Mainfueltype	Categorical	Primary fuel used for heating
	DWtype	Categorical	Dwelling type (detached, terraced, flat, etc.)
	DWage	Categorical	Age band of the dwelling
Socio-economic	WallType	Categorical	Wall construction type
	FloorArea	Categorical	Total floor area band
	Boiler	Categorical	Type of boiler installed
	HHincome	Numerical	Annual gross household income
	HHsick	Categorical	Household contains member with long-term illness/ disability
	WorkingStatus	Categorical	Employment status of household reference person
	Ageold	Categorical	Age band of the oldest household member

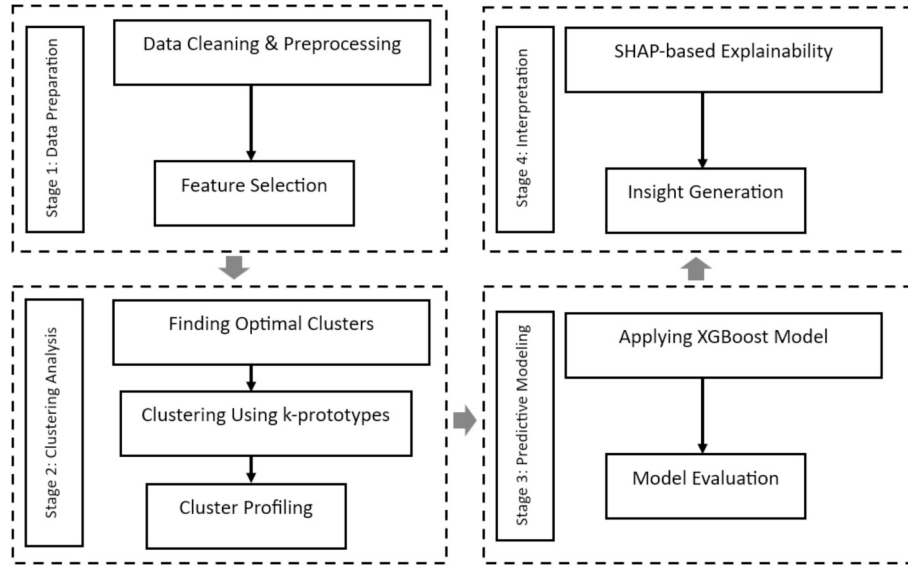


Fig. 1. The flowchart methodology steps.

3.2.2. Clustering *k*-prototypes

Clustering analysis groups households into profiles that are internally homogeneous but distinct from other groups [45]. Among the various algorithms available, *k*-prototypes is particularly suitable for mixed datasets, as it combines the strengths of *k*-means (numerical data) and *k*-modes (categorical data) [46,47]. The algorithm minimizes a joint cost function that balances squared Euclidian distances for numerical features with a matching coefficient for categorical features. The cost function utilized is defined as follows [48]:

$$X_i = [x_{i,1}^n, x_{i,2}^n, \dots, x_{i,p}^n, x_{i,p+1}^c, x_{i,p+2}^c, \dots, x_{i,m}^c], \quad (2)$$

$$E = \sum_{l=1}^k \sum_{i=1}^t w_{i,l} d(X_i, Q_l), \quad (3)$$

where $w_{i,l} = 1$ if household i is assigned to cluster l and 0 otherwise, and Q_l is the prototype (centroid) of cluster l . The full derivation, including decomposition into numerical and categorical terms, is provided in Appendix A.3 (Eqs. (A1)–(A3)) [49].

In this study, the *k*-prototypes algorithm was applied to the 11,974 households' characteristics. The number of clusters (k) was chosen using the “elbow method”, which evaluates reductions in the cost function as k increases and identifies a turning point or “elbow”, at which point it levels off. This approach ensures that resulting clusters remain parsimonious and interpretable.

After identifying the optimal number of clusters with the *k*-prototypes algorithm, we characterized each cluster in terms of household and dwelling attributes. To support this step, we used the visualization tool Microsoft Power BI (Business Intelligence) software that enabled clear inspection of variable distributions and facilitated detection of patterns within clusters. [50–52]. This ensured that profiles were not only statistically defined but also interpretable in socio-economic, housing, and energy-cost characteristics (see subsection 4.1 below).

3.2.3. XGBoost algorithm and model evaluation

XGBoost is an ensemble algorithm based on gradient boosting principles and is a supervised learning boosting technique that originates in tree-based models [53]. This algorithm combines predictions of multiple ‘weak’ classifiers through a sequential training process to form a ‘strong’ classifier, while overfitting is mitigated through regularization. Parallel and distributed computing further accelerate the learning process, making XGBoost particularly efficient [53].

In this study, the XGBoost method was selected to predict the fuel poverty status within clusters for four main reasons. First, it is among the most widely used boosting tree algorithms and consistently delivers high prediction performance. Second, it effectively captures complex, non-linear relationships between variables [54,55]. Third, given the large sample size ($N = 11,974$), it is computationally efficient compared to deep learning approaches [56]. Finally, XGBoost provides partial interpretability through feature importance measures, allowing stakeholders to understand the drivers behind predictions [57]. The computational process of XGBoost is schematically illustrated in Appendix A.4 (Fig. A2). The full mathematical formulation of ensemble predictions, objective function, and regularization terms is provided in Appendix A.4 (Eqs. (A4)–(A8)). The XGBoost method can take both continuous and discrete features as inputs, but it requires the output feature to be discrete, including binary variables. In this article, the XGBoost method was implemented using Python 3.11.

As the ML models grow in complexity, their internal logic often becomes difficult to interpret, raising concerns about the credibility of predictions in high-stakes applications such as household vulnerability analysis [58,59]. High prediction accuracy alone is insufficient if models cannot explain why certain households are classified as fuel-poor. To address this, we apply SHAP, a method introduced by Lundberg and Lee [60] that builds on cooperative game theory to decompose each prediction into feature-level contributions. In this study, SHAP highlights which socio-economic, housing, and energy cost variables most strongly drive classification outcomes, providing transparent insights into the mechanisms of fuel poverty risk.

We evaluated the predictive performance of XGBoost using 10-fold cross-validation, ensuring that each subset of the data served once as a test set and nine times as part of the training set [61]. To capture different aspects of classification quality, we report standard metrics: accuracy (overall correctness), precision (share of predicted fuel-poor households that are truly fuel-poor), recall or sensitivity (ability to correctly identify all fuel-poor households), F1-score (harmonic mean of precision and recall), and ROC-AUC (ability to discriminate between fuel-poor and non-fuel-poor households) [62–64].

4. Results and discussions

This study demonstrates the effectiveness of the *k*-prototypes clustering approach in identifying distinct household vulnerability profiles within the entire EHS dataset. Four XGBoost models were developed:



Fig. 2. Distribution of variables within clusters, variables 1–12.
Distribution of variables within clusters, variables 13–18.

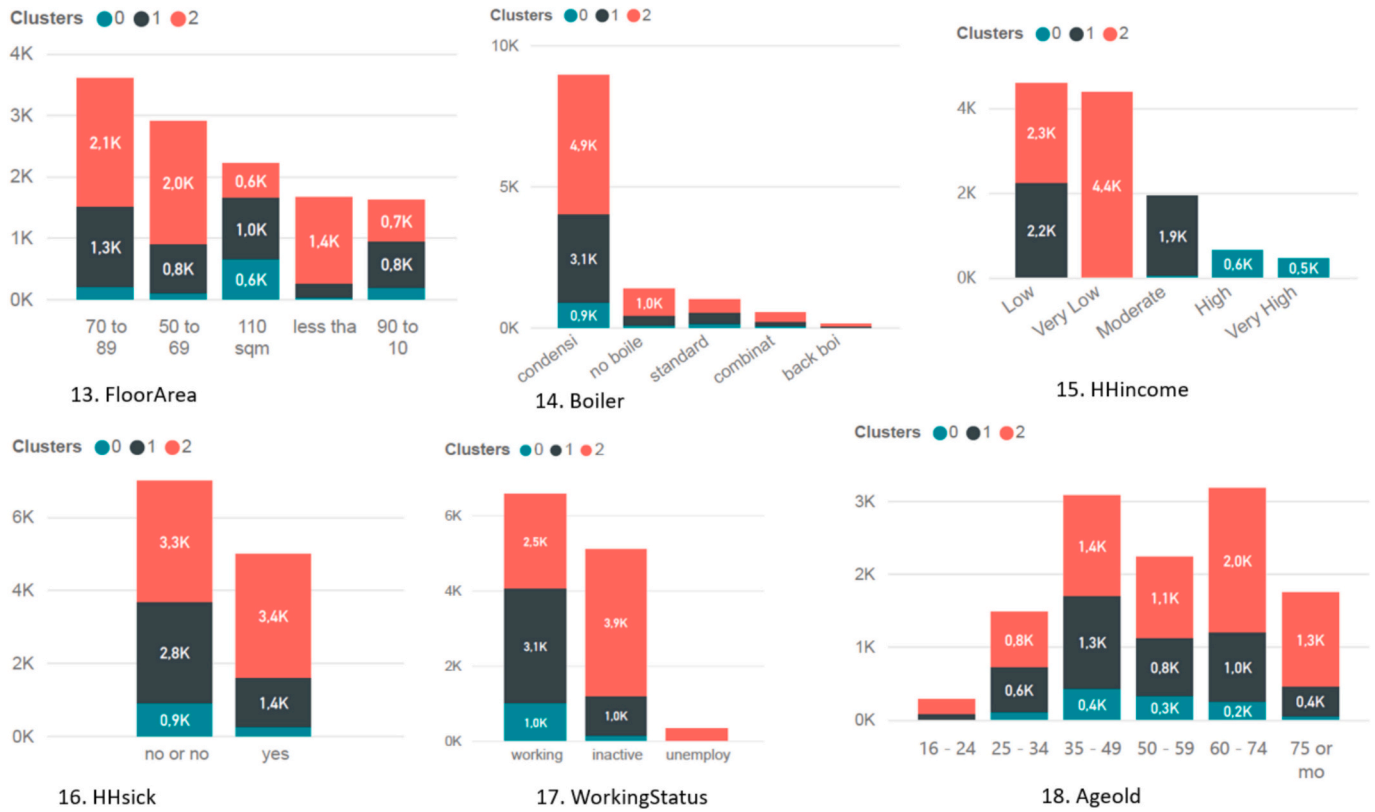


Fig. 2. (continued).

one trained on the entire dataset, and three trained separately on the identified clusters using the 18 input features listed in Table 1. The EHS provides a binary classification of the fuel-poverty status under the statutory LILEE definition, used here solely as the outcome variable for predictive modeling and performance evaluation. Vulnerability was therefore not imposed ex ante but emerged endogenously as a defining characteristic of one cluster (for details see Section 4.1).

This framework integrates clustering and predictive modeling to uncover heterogeneous vulnerability patterns that the static national indicators cannot capture. To highlight the advantages of this approach, the predictive performance of the overall model was compared with that of the cluster-based models. Subsequently, the SHAP method was applied to assess and compare the influence of contributing factors on fuel poverty within each cluster.

4.1. Cluster analysis and profiling

Before applying the k-prototypes algorithm, it is necessary to determine the optimal number of clusters (k -value). Among various established methods in the literature, this research employed the elbow method, as it aligns well with the aim of this study. The elbow method is a well-established technique for finding the optimal number of clusters in a given dataset [65]. Its fundamental concept is to start setting an initial optimal cluster number, typically $k = 2$, and gradually increasing k step by step until reaching the maximum specified for estimating the potential optimal cluster number. Finally, the method identifies the optimal cluster number (k^*) by observing a distinct “elbow” point in the cost curve. The optimal value (k^*) is distinguished by a rapid decrease in cost before reaching k , followed by a much slower rate of decrease after surpassing k , leading to a noticeable curve bend. The corresponding cost curve and identified elbow point are shown in Appendix A.5 (Fig. A3), confirming three clusters as the optimal configuration. The research identifies three clusters across the entire dataset by applying the k-prototypes algorithm. A visualization of the clusters' distribution is

provided in Appendix A.5 (Fig. A4).

The clusters are labeled from 0 to 2, following the default indexing convention of the scikit-learn implementation in Python; this indexing is purely nominal and has no impact on the composition, interpretation, or statistical properties of the clusters. For visualization purposes, the x-axis and y-axis correspond to the first two principal components, used to reduce the dimensionality of the dataset. One of the clusters is dominated by older, low-income, and economically inactive households, identified as the vulnerable group; detailed explanations of each cluster are provided in the following. Cluster analysis plays a key role in predicting household energy poverty by enabling the identification of groups with similar socio-economic, housing, and energy cost characteristics. By uncovering the main features that contribute to fuel poverty, this approach aids policymakers in developing effective interventions and policies for the targeted group.

To further explore intra-cluster patterns, the model outputs were investigated using Microsoft's Power BI software. The numerical data were transformed into categorical formats to facilitate comparisons with the existing categorical dataset. For example, “SpaceHeatingCost” is originally numerical data representing actual cost in monetary values. We transformed this numerical data into categorical formats by defining specific ranges and assigning descriptive categories such as “Very Low”, “Low”, “Moderate”, “High”, “Very High”, and “Entirely High” based on England's tariff structure. Fig. 2 presents the distribution of each variable across the three clusters.

Based on the patterns shown in Fig. 2 and its continued panels, notable characteristics of each cluster can be interpreted as follows:

Cluster 0 – Moderate-cost households at potential risk

Households in this group exhibit moderate cooking and water-heating costs and mainly occupy detached dwellings with an average floor area of 110 m². Most report a relatively good household health situation and maintain balanced energy consumption. Although currently not fuel-poor, these households could become vulnerable under rising energy prices or future income shocks.

Cluster 1 – Younger, employed households with high appliance and lighting costs

Cluster 1 comprises households with a relatively younger demographic, aged between 35 and 49 years. These households live in semi-detached dwellings with floor areas ranging from between 70 and 89 m², and are predominantly owner-occupied. Cooking, space heating, and water-heating costs are generally moderate; however, expenditures on lighting and electrical appliances are noticeably higher than in the other clusters. The cluster is mainly composed of employed couples without children, many of whom report a reasonably good household health. Their energy consumption pattern reflects comfort- and lifestyle-driven electricity use, associated with higher appliance ownership and more intensive daily usage, rather than with basic heating needs.

Cluster 2 – Low-income, economically inactive older households (vulnerable group)

Cluster 2 consists of households primarily aged between 60 and 74 years. These households typically incur moderate cooking costs and live in purpose-built dwellings equipped with condensing boilers, with most dwellings built after 1945. This cluster is characterized by being vulnerable, evident through very low household incomes. They maintain relatively low expenses for space heating, possibly indicating challenges in affording adequate warmth. However, notably high expenses for lighting and appliances indicate that a substantial portion of their income is allocated to electricity. The common household composition in this cluster is single occupancy, which might indicate limited financial resources. The majority of individuals are economically inactive, reinforcing their economic vulnerability and heightened risk of fuel poverty.

4.2. XGBoost modeling

In this section, in order to estimate the XGBoost prediction performance for the clustering step, the 10-fold cross-validation method is applied. The accuracy rate for each fold (subset) in Table 2 is shown, where k represents the k th verification of the 10-fold cross-validation technique.

The average accuracy rate of the XGBoost prediction performance equals to 0.9116, indicating that the XGBoost modeling has a good prediction performance and can be used to assess the accuracy of clustering results.

Then, in the next step, the predicting performances of the models were assessed based on precision, sensitivity or recall, F1-score, and ROC-AUC score (see notes in Table 3). This research shows that the k-prototypes cluster-based models have a high accuracy rate at predicting fuel poverty. All the models were trained and tested on 75 % and 25 % of the observations in the dataset. Then, as shown in Table 3 the entire dataset (EDS) model was compared with the k-prototypes clustering-based models in terms of accuracy, validation, precision, recall, F1-score, and ROC-AUC score.

Table 3 illustrates the predictive performance of distinct clusters in forecasting fuel poverty risk. Cluster 0 demonstrates respectable accuracy, precision, recall, and F1-Score, albeit with a comparatively lower ROC-AUC score. Cluster 1 exhibits overall improved metrics compared to Cluster 0. Particularly noteworthy is the outstanding performance of Cluster 2, showcasing superior predictive capabilities across all evaluated metrics. Notably, the utilization of the k-prototypes clustering

Table 2
Ten-fold cross-validation.

k	Accuracy rate	k	Accuracy rate
1	0.9090	6	0.9140
2	0.9132	7	0.9315
3	0.9057	8	0.9248
4	0.9132	9	0.9081
5	0.8972	10	0.8989

Table 3

Predicting performance evaluation of cluster-based models.

Cluster	Accuracy	Validation	Precision	Recall	F1-Score ^a	ROC-AUC ^b
0	0.9012	0.8405	0.8405	0.8405	0.8405	0.5541
1	0.9520	0.8518	0.8518	0.8518	0.8518	0.6287
2	0.9864	0.9031	0.9031	0.9031	0.9031	0.7727
EDS ^c	0.9916	0.9119	0.9119	0.9119	0.9119	0.8023

^a F1-Score is a metric that combines precision and recall. It considers both false positives and false negatives.

^b ROC-AUC (Receiver Operating Characteristic – Area Under the Curve).

^c EDS- Entire Dataset.

method significantly enhanced predictive performance, as evidenced by the remarkable results achieved for Cluster 2. However, while the individual clusters exhibited strong predictive potential, the model trained on the EDS emerged as the top performer across all evaluation criteria. This emphasizes the collective strength and superior predictive capacity of the clustering-based model in effectively identifying households at risk of fuel poverty.

4.3. Feature analysis

In this section, the feature analysis conducted is described. The utilization of the SHAP (Shapley additive explanations) approach for feature analysis reveals that applying k-prototypes clustering analysis is a useful step in discovering latent patterns in clusters. Fig. 3 shows the SHAP value of features for the predicting fuel-poor households within cluster 0, cluster 1, and cluster 2, respectively.

The suffixes appended to the label of each feature represent the unique values found within that feature. The x-axis in the figure shows how a feature influences the model's output, while the right-hand side vertical axis illustrates the feature's value as either low (blue) or high (red). On the one hand, when a specific value of a feature leads to higher SHAP values, it highlights the risk of fuel poverty. On the other hand, if a feature value results in lower SHAP values, it decreases the risk of fuel poverty. The most important features revealed in Fig. 3 are summarized in Table 4. Table 4 outlines related features for predicting fuel poverty within different clusters of households as “fuel-poor” and “not fuel-poor”. Unlike econometric models, which estimate average effects across the population, SHAP highlights how feature importance varies across different household clusters. In Cluster 0, the most influential predictors are water heating costs, floor area, and tenure type. Households with higher water heating costs and smaller floor areas face elevated risk, as do those in social or less secure tenure arrangements. Employment status (inactive) and mid-terrace dwelling types also contribute to vulnerability. These results suggest that affordability challenges in this group are shaped by structural housing and tenure conditions, rather than extreme energy burdens.

Econometric studies using the EHS dataset have likewise shown that tenure and dwelling type are significant predictors of fuel poverty at the national level [31]. However, our SHAP results reveal that these factors are especially decisive in this cluster, whereas they matter less in the others. This cluster-specific perspective thus adds explanatory depth that static econometric models cannot provide.

For Cluster 1, the SHAP plots highlight space heating costs, lighting/appliance costs, and household composition variables as the dominant predictors. Even though many households in this cluster are employed, high electricity use—particularly for lighting and appliances—substantially raises the likelihood of fuel poverty. Couples without children and multi-person households are especially affected, underscoring the role of consumption practices alongside structural variables. These findings align with conceptual frameworks that identify occupant behavior, such as energy use patterns, as critical fuel poverty determinants [66]. However, our SHAP results suggest that electricity-driven consumption is a dominant vulnerability pathway primarily for certain households,

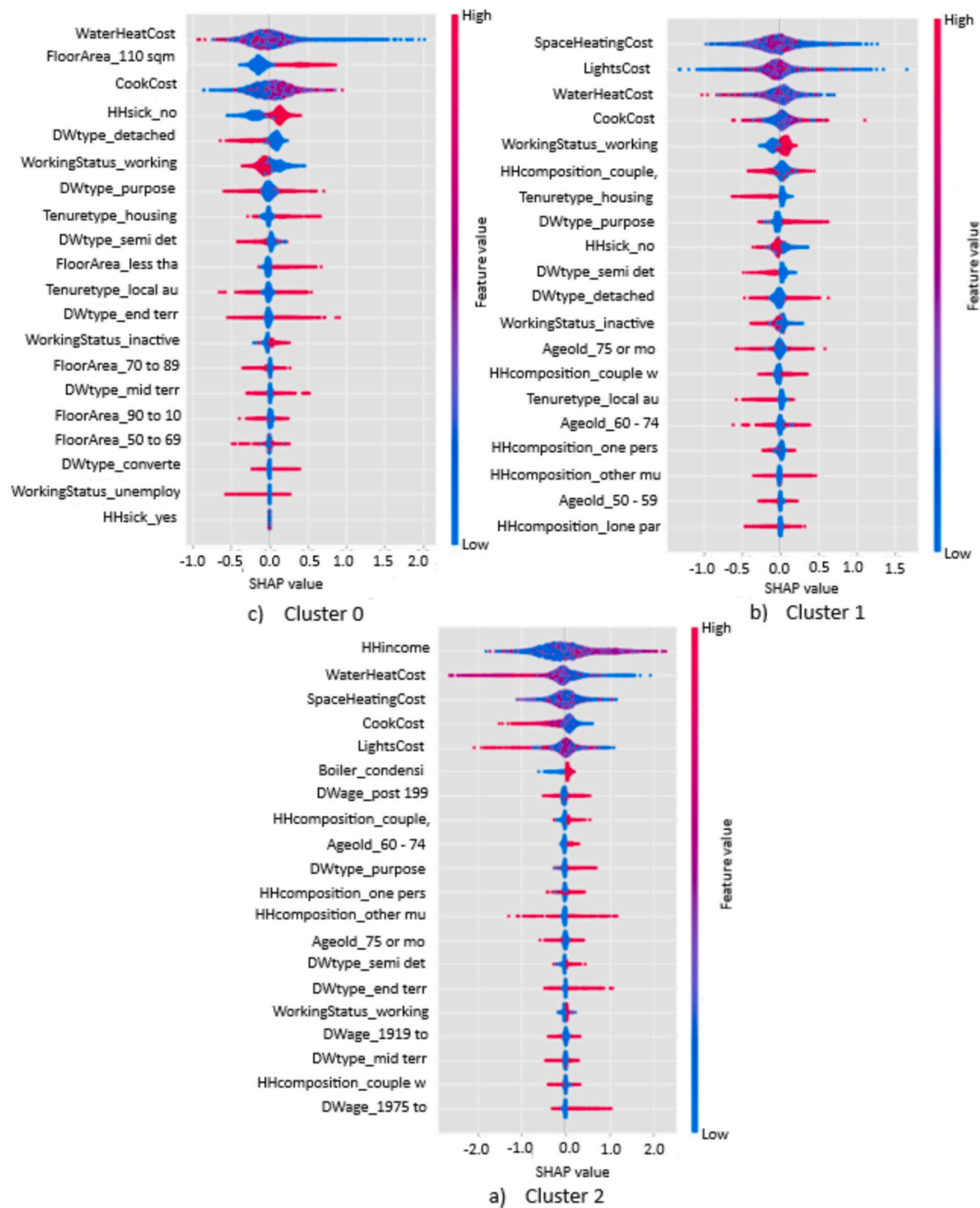


Fig. 3. Summary plot of important features in the three clusters-based model.

rather than a universal driver across all households.

Cluster 2 is characterized by low income, older age (60–74 years), and single or couple household composition, which together sharply increase fuel poverty risk. Dwelling age (post-1975 and post-1990 stock) and end-terrace dwellings also contribute, likely reflecting inefficiencies in the housing stock. This cluster captures the structural disadvantage of older, economically inactive, low-income households—a group repeatedly identified as fuel-poor in UK studies.

Taken together, the cluster-specific SHAP analyses confirm and extend findings from the UK fuel poverty literature. Structural factors such as tenure and dwelling type, which have been long recognized in econometric models [16,24], remain crucial for certain groups but not for others. Recent reports emphasize that affordability pressures increasingly extend to electricity-related uses, a trend our Cluster 1

results capture explicitly [40]. At the same time, the entrenched vulnerability of low-income, older households documented in earlier work [16,67,68] is strongly reproduced in Cluster 2.

5. Conclusion and policy implications

This study advances fuel poverty research by applying a clustering-prediction-interpretation framework to assess fuel poverty risk in England using the 2019 EHS dataset. By integrating k-prototypes clustering with XGBoost modeling and SHAP interpretability, the analysis identified three distinct household vulnerability profiles and revealed the socio-economic, housing, and energy-related characteristics that most strongly shape fuel poverty risk. Cluster 2, comprising low-income and economically inactive older households, emerged as the most

Table 4
Important features of predicting fuel poverty in the three clusters identified.

Fuel poverty prediction	Input features	Cluster 0	Cluster 1	Cluster 2
Fuel-poor	WaterHeatCost	✓		
	FloorArea-110	✓		
	Tenuretype_housing	✓		
	FloorArea_less	✓		
	WorkingStatus_inactive	✓		
	DWtype_mid terr	✓		
	DWtype_convert	✓		
	SpaceHeatingCost		✓	
	DWtype_purpose		✓	✓
	HHcomposition_couple w		✓	
	HHcomposition_other mu		✓	✓
	HHincome			✓
	DWage-post 199			✓
	HHcomposition_couple,			✓
	Ageold_60.74			✓
	DWtype_end terr			✓
	DWage_1975 to			✓
Non-fuel poor	CookCost	✓		
	HHsick_no	✓		
	DWtype-detached	✓		
	DWtype_semi det	✓	✓	
	FloorArea_70 to 89	✓		
	FloorArea_90 to 10	✓		
	FloorArea_50 to 69	✓		
	Tenuretype_housing		✓	
	Tenuretype_local au		✓	
	HHcomposition_lone par		✓	
	whatheatCost			✓
	CookCost			✓

vulnerable, while Clusters 0 and 1 captured households with moderate and higher energy costs, respectively. These results demonstrate the value of combining clustering and predictive modeling to uncover heterogeneity that static fuel poverty indicators, such as LILEE, often mask.

The study underscores that the effective alleviation of fuel poverty requires differentiated strategies tailored to the specific vulnerability pathways revealed by clustering. For Cluster 2 (economically vulnerable, older, and inactive households), targeted support is essential given their low incomes and limited capacity to invest in energy efficiency. Policy measures should therefore prioritize direct financial transfers (e. g., expansions of the Warm Home Discount or targeted winter fuel payments), subsidized retrofitting and insulation programs (grants for condensing boilers, wall and loft insulation upgrades), and enhanced social protection mechanisms for households with elderly or disabled members. Outreach programs could ensure that these groups, who are often underrepresented in existing schemes, are actively enrolled. For Cluster 1 (younger, employed households with high appliance and lighting costs), affordability challenges stem mainly from electricity-intensive consumption rather than heating. Policies should therefore emphasize appliance and lighting efficiency. This can be achieved through stronger minimum energy performance standards, rebates or low-interest loans for efficient appliances, and targeted behavioral campaigns to shift consumption habits. Workplace-linked efficiency programs and employer partnerships could also be effective, given the higher employment rates of this group. For Cluster 0 (moderate-cost households at potential risk), preventive measures are most relevant.

Although currently less vulnerable, these households may face fuel poverty under future price shocks. Light-touch interventions such as smart meter rollouts, draught-proofing, and minor retrofits can build resilience. Awareness campaigns to promote energy-saving practices and eligibility checks for early intervention schemes would help ensure that households in this cluster do not slip into fuel poverty during periods of sustained price volatility.

More broadly, the findings suggest that uniform national schemes are insufficient. Instead, tailoring interventions to cluster-specific profiles can enhance efficiency in resource allocation, ensuring that support reaches those most in need while preventing future vulnerability. By leveraging predictive modeling alongside clustering, policymakers gain a roadmap that identifies not only who is at risk but also why, enabling more cost-effective and socially equitable interventions. This approach helps prioritize resources where they generate the greatest welfare gains, align assistance with the underlying drivers of vulnerability, and strengthen the long-term resilience of households against future energy price shocks or economic downturns.

While the methodology provides robust insights, several limitations and design considerations should be acknowledged. The analysis relies on the 2019 EHS dataset, the latest publicly available microdata with household-level fuel poverty indicators. This constrains our ability to capture post-2022 price shocks, though structural drivers such as tenure, housing stock, and heating systems evolve gradually, ensuring continued relevance of the findings. Future releases of disaggregated microdata would allow assessment of crisis-specific dynamics and further validation of the framework. Methodologically, the study is cross-sectional and observational; it identifies vulnerability profiles but does not permit causal inference or dynamic analysis over time. To strengthen rigor and validity, we combined clustering with predictive modeling, applied cross-validation to test model performance, and used SHAP interpretability to avoid black-box outcomes. Even so, SHAP values do not represent marginal causal effects and should rather be read as indicators of relative importance. The study is also limited to England; while the framework is transferable to other national contexts, vulnerability profiles are shaped by country-specific housing stocks, energy systems, and policy regimes. Extending the analysis to longitudinal datasets and testing alternative ML methods would allow stronger validation and improve generalizability. Acknowledging these design choices reveals both the strengths and the limits of our framework, and supports recent calls for greater methodological transparency in energy and social science research.

CRediT authorship contribution statement

Rahil Dejkam: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Reinhard Madlener:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

A.1. Descriptive statistics of variables

Table A1

Descriptive statistics of the variables used in the analysis.

Category	Variables	Type	Description	Descriptive Statistics
Energy costs	SpaceHeatingCost	Numerical	Annual expenditure on space heating (£)	Mean = 540, SD = 319, Min = 0, Max = 4468
	LightsCost	Numerical	Annual expenditure on lighting/appliances (£)	Mean = 529, SD = 192, Min = 0, Max = 2261
	CookCost	Numerical	Annual expenditure on cooking (£)	Mean = 49, SD = 12, Min = 0, Max = 129
	WaterHeatCost	Numerical	Annual expenditure on water heating (£)	Mean = 138, SD = 78, Min = 0, Max = 828
Housing Characteristics	Tenuretype	Categorical	Dwelling tenure	Owner-occupied 44.8 %; Housing association 20.8 %; Private rental 20.2 %; Local authority 14.2 %
	HHcomposition	Categorical	Household composition	Couple 31.1 %; Single 31.0 %; Couple w/children 20.8 %; Lone parent 8.7 %; Other 8.5 %
	Loftins	Categorical	Presence of loft insulation	≥125 mm 56.2 %; <125 mm 27.5 %; Not applicable 16.3 %
	FuelPoverty	Categorical	Fuel poverty flag (LILEE indicator)	Fuel poor 15.6 %; Not fuel poor 84.4 %
	Mainfueltype	Categorical	Primary fuel used for heating	Gas 85.2 %; Electricity 8.7 %; Other 6.1 %
	DWtype	Categorical	Dwelling type (detached, terraced, flat, etc.)	Semi-detached 26.5 %; Purpose-built flat 21.7 %; Mid-terrace 19.7 %; Detached 16.2 %; End-terrace 12.2 %; Converted flat 3.8 %
	DWage	Categorical	Age band of the dwelling	1945–64: 21.7 %; Post-1990: 18.0 %; 1965–80: 14.5 %; 1919–44: 13.3 %; Other bands <10 %
	WallType	Categorical	Wall construction type	Cavity insulated 49.7 %; Solid uninsulated 22.6 %; Cavity uninsulated 21.6 %; Other 6.1 %
	FloorArea	Categorical	Total floor area band	Fuel poor (LILEE): Yes (15.6 %), No (84.4 %)
	Boiler	Categorical	Type of boiler installed	Condensing (74.6 %), none (11.4 %), standard (8.4 %), combination/back (5.6 %)
Socio-economic	HHincome	Numerical	Annual gross household income	Mean = 30,784; SD = 20,532; Min = 0; Max = 100,000
	HHsick	Categorical	Household contains member with long-term illness/disability	Yes 41.6 %; No 58.4 %
	WorkingStatus	Categorical	Employment status of household reference person	Working 54.8 %; Inactive 42.4 %; Unemployed 2.8 %
	Ageold	Categorical	Age band of the oldest household member	35–49: 25.6 %; 60–74: 26.5 %; 50–59: 18.6 %; 75+: 14.5 %; 25–34: 12.4 %; 16–24: 2.3 %

Note: Continuous variables are reported as mean, standard deviation (SD), minimum, and maximum. Categorical variables are reported as percentages of households. $N = 11,974$ households (English Housing Survey: Fuel Poverty Dataset 2019).

A.2. Correlation heatmaps for feature selection

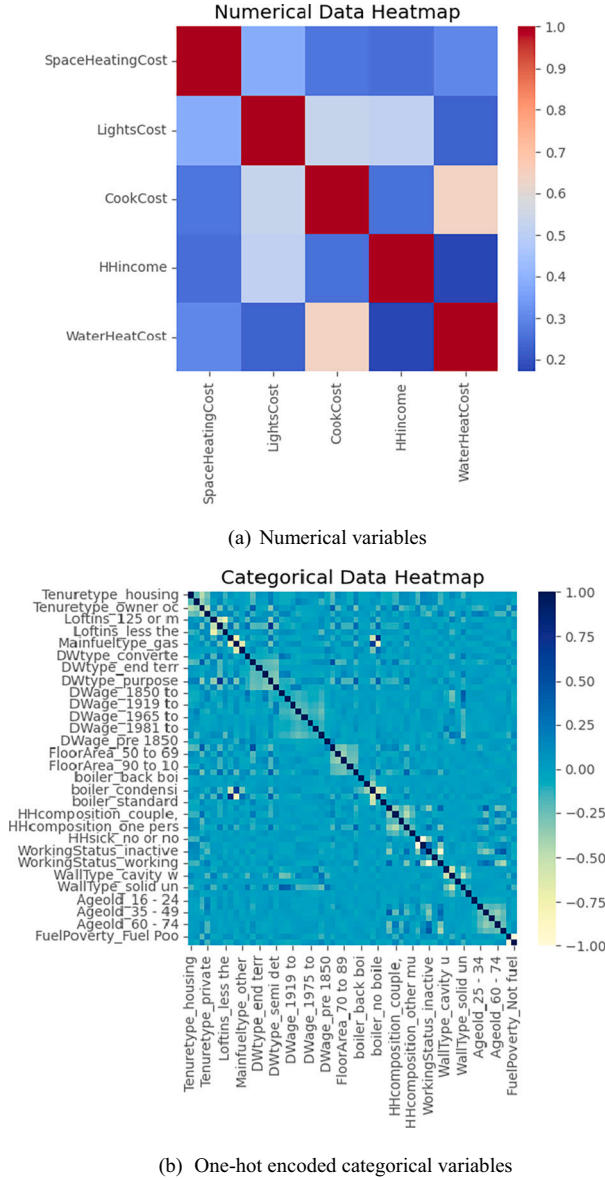


Fig. A1. Pearson correlation coefficient heatmaps for (a) numerical variables and (b) one-hot encoded categorical variables. The color scale ranges from -1 (perfect negative correlation) to $+1$ (perfect positive correlation).

A.3. *k*-prototypes clustering: technical details

For completeness, this appendix presents the formal specification of the *k*-prototypes algorithm used in Section 3.2.3.

A household i is represented by a mixed feature vector:

$$X_i = [x_{i,1}^n, x_{i,2}^n, \dots, x_{i,p}^n, x_{i,p+1}^c, x_{i,p+2}^c, \dots, x_{i,m}^c], \quad (\text{A1})$$

where the first p features are numerical and the remaining $(m-p)$ are categorical.

The dissimilarity measure $d(\cdot)$ combines squared Euclidean distances for numerical features with a matching dissimilarity for categorical features:

$$d(X_i, Q_l) = \sum_{j=1}^p (x_{ij}^n - q_{lj}^n)^2 + \gamma_l \sum_{j=p+1}^m \delta(x_{ij}^c, q_{lj}^c), \quad (\text{A2})$$

where $\delta(\cdot)$ equals 0 if the categorical values match and 1 otherwise, and γ_l is a weighting parameter that balances the influence of categorical versus numerical features. The total cost can thus be written as:

$$E_l = \sum_{i=1}^t y_{il} \sum_{j=1}^{m_n} (x_{ij}^n - q_{lj}^n)^2 + \gamma_l \sum_{i=1}^t y_{il} \sum_{j=1}^{m_c} \delta(x_{ij}^c, q_{lj}^c) = E_l^n + E_l^c, \quad (\text{A3})$$

where y_{il} indicates assignment of observation i to cluster l .

A.4. Mathematical details of XGBoost

The additive model is defined as:

$$\hat{y}_i^{(t)} = \sum_{k=1}^t f_k(x_i) = \hat{y}_i^{(t-1)} + f_t(x_i), \quad (\text{A4})$$

where the number of base tree models is t , and each tree model $f_t(x_i)$ contributes leaves, which are the terminal nodes where predictions are made. The newly generated tree model is $f_t(x_i)$, the previously generated tree model is $\hat{y}_i^{(t-1)}$, and the final tree model after adding the new tree is $\hat{y}_i^{(t)}$. Both depth and the total number of trees are crucial factors for finding the optimal XGBoost algorithm. This is achieved by minimizing a regularized loss function, expressed as follows:

$$Y^{(t)} = \sum_{i=1}^t L(y_i, \hat{y}_i^{(t)}) + \sum_{i=1}^t \Omega(f_i), \quad (\text{A5})$$

where the actual value is y_i , the predicted value is $\hat{y}_i^{(t)}$, the loss function is denoted as $L(y_i, \hat{y}_i^{(t)})$, and the regularization parameter is $\Omega(f_i)$ [69].

Substituting Eq. (A4) into Eq. (A5) yields:

$$Y^{(t)} = \sum_{i=1}^t L(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_i) + \text{constant}. \quad (\text{A6})$$

which, after a second-order Taylor expansion, becomes:

$$Y^{(t)} = \sum_{i=1}^t L \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_i), \quad (\text{A7})$$

where $g_i = \partial_{y_i^{(t-1)}} l(y_i, y_i^{(t-1)})$ and $h_i = \partial_{y_i^{(t-1)}}^2 l(y_i, y_i^{(t-1)})$ are the first-order and second-order gradient statistics on the loss function.

The regularization term penalizes model complexity:

$$\Omega(f_i) = \gamma T + \frac{1}{2} \lambda \|\omega\|^2, \quad (\text{A8})$$

where T is the number of leaves, ω is the weight of the leaves, λ and γ are regularization coefficients (default values: $\lambda = 1$ and $\gamma = 0$).

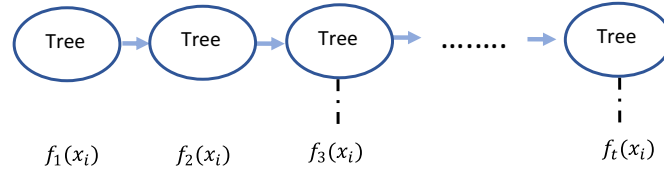


Fig. A2. Outline of the XGBoost algorithm.

A.5. Cluster analysis and profiling

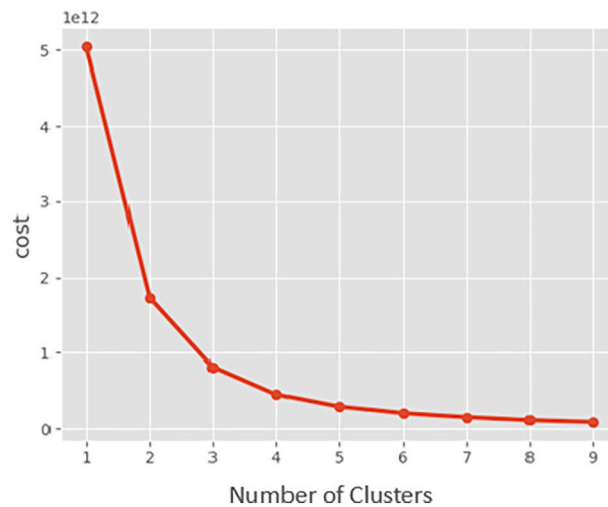


Fig. A3. Optimal number of clusters determined by the elbow method.

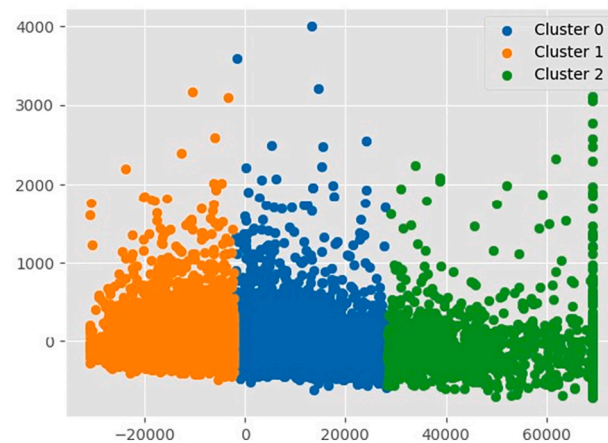


Fig. A4. k-prototypes clustering of English households into three clusters.

Data availability

Data will be made available on request.

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